

AI & ML Lab – Final Lab Assignment Links

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CNN Image Classification – Geometric Shapes

Repository and Notebooks

1. **GitHub Profile:**

<https://github.com/ahsanjust/>

2. **Final Lab Standalone Assignment Repository:**

<https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification>

3. **CNN – GitHub Notebook:**

https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/220119_CNN.ipynb

4. **Google Colab Notebook (GitHub-Native, Direct Link):**

https://colab.research.google.com/github/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/220119_CNN.ipynb

5. **Repository README:**

<https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/readme.md>

Performance Summary – CNN (Standard Test Set)

Class	Precision	Recall	F1-Score	Support
Circles	1.0000	0.9000	0.9474	30
Squares	1.0000	1.0000	1.0000	30
Triangles	0.9091	1.0000	0.9524	30
Overall Accuracy		0.9556 (95.56%)		

Architecture & Workflow Summary

- **Model:** 2-Layer CNN (PyTorch) – Conv2d(3→32) → ReLU → MaxPool2d(2) → Conv2d(32→64) → ReLU → MaxPool2d(2) → Flatten → Linear(16384→128) → ReLU → Linear(128→3)
- **Input Preprocessing:** 64×64 RGB images with single shared transform: Resize → ToTensor → Normalize($\mu = 0.5, \sigma = 0.5$)
- **Training Setup:** CrossEntropyLoss, Adam optimizer (lr = 0.001), batch size 64, 25 epochs
- **Dataset:** Geometric Shapes (Option 5) – train 543, validation 180, test 90, plus 15 custom smartphone photos

- **Real-World Accuracy:** 12 / 15 (80.0%) on custom smartphone photos (Circles: 4/5, Squares: 3/5, Triangles: 5/5) – circle4.jpg correctly classified at 98.9% confidence
- **Zero-Manual-Upload Automation:** Clones repo automatically in Google Colab, loads weights from `model/220119.pth`, evaluates test set, and predicts on custom photos automatically upon clicking “Run All”.